

OSINT Domain Reconnaissance using Python

Python Program:

```
import socket

domains = [
    "www.google.com",
    "www.python.org",
    "notarealdomain12345.com"
]

print("OSINT Domain Reconnaissance")

for d in domains:
    try:
        ip = socket.gethostbyname(d)
        print(f"{d:30s} -> {ip}")
    except socket.gaierror:
        print(f"{d:30s} -> Could not resolve (invalid/unreachable)")
```

Sample Output:

OSINT Domain Reconnaissance

```
www.google.com           -> 142.251.151.119
www.python.org           -> 151.101.0.223
notarealdomain12345.com  -> Could not resolve (invalid/unreachable)
```

Explanation:

This Python program demonstrates a basic Open Source Intelligence (OSINT) domain reconnaissance technique. It uses the **socket** module to perform DNS lookups and resolve domain names into their corresponding IP addresses. If a domain exists, its IP address is displayed. If the domain does not exist or cannot be reached, the program catches the exception and reports that the domain could not be resolved. This technique is commonly used during the information gathering phase of cybersecurity assessments and digital forensic investigations.